



INTERNSHIP REPORT

AWS Data Engineering Virtual Internship

Submitted to Mrs. Surabhi Purwar, Assistant Professor
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CS-23411630 | B.Tech CSE | Batch 2023–27

Internship Completion Snapshot

Official proof of successful completion, awarded through EduSkills Foundation



8 Weeks

Structured Virtual Program



AWS Academy

Curriculum Provider



Certified

Virtual Internship Certificate



EduSkills

Foundation, with AICTE

What the Certificate Confirms

Namita Sharma, a student of IILM University, Greater Noida, successfully completed the 8-week Data Engineering Virtual Internship (June – August 2026), with the curriculum delivered by AWS Academy through the AICTE–EduSkills Virtual Internship Program.

Introduction

Why data engineering and cloud computing matter today

Modern organizations generate massive volumes of data from applications, websites, transactions, sensors, and other sources every day.

Data Engineering involves collecting, storing, processing, transforming, and preparing this data for analysis.

Cloud Computing makes these processes scalable and flexible by delivering computing and storage resources over the internet.

This internship built a foundational understanding of cloud-based data engineering using Amazon Web Services (AWS), connecting academic concepts with industry-oriented technology.



AWS — Amazon Web Services

A leading cloud platform offering services for computing, storage, databases, data processing, security, and analytics
— the backbone of this internship's learning path.

Organization Profile

EduSkills Foundation — bridging academia and industry



EduSkills Foundation

A skill-development organization focused on providing industry-oriented learning and virtual internship opportunities to students, reducing the gap between academic education and industry requirements.

Focus Areas

Cloud Computing • AI & ML • Data Science
Data Engineering • Cybersecurity

Role in This Internship



Delivered the AWS Data Engineering curriculum via the AWS Academy platform.



Provided a week-wise structured learning plan with weekly assessments.



Issued the AICTE–EduSkills Virtual Internship Completion Certificate.

Problem Statement

Why organizations need scalable, reliable data engineering solutions

Modern organizations generate huge amounts of data in varied formats from many sources. Managing, storing, processing, and analyzing this data efficiently is difficult with traditional systems — data often needs cleaning and transformation before it can be used.



The internship focused on understanding how AWS cloud services handle data storage, processing, transformation, integration, and analytics — enabling scalable, reliable, cost-effective, and data-driven decision-making.

Project Objectives

Ten goals guiding the internship's learning path

- 1 Understand the fundamentals of cloud computing and AWS
- 2 Grasp the basic concepts of Data Engineering
- 3 Learn cloud-based data storage and processing
- 4 Understand ETL (Extract, Transform, Load) processes
- 5 Understand data pipelines and data integration
- 6 Gain knowledge of AWS services used in data engineering
- 7 Understand how processed data supports analytics
- 8 Build industry-oriented technical knowledge & exposure
- 9 Connect academic concepts with real-world technologies
- 10 Develop a foundation for further AWS & Data Engineering learning

Technologies & Tools Used

The AWS services and languages studied during training



AWS

Cloud platform foundation



Amazon S3

Object storage & data lake



AWS Glue

Data integration & ETL



Amazon Redshift

Data warehousing



Amazon Athena

Query data in S3 via SQL



SQL

Structured data queries



Python

Programming & data processing

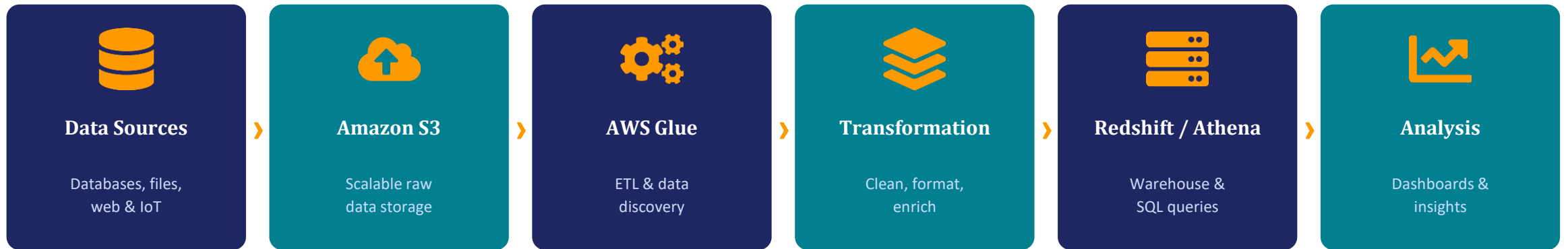


AWS IAM

Identity & access management

System Architecture

How AWS services combine into an end-to-end data workflow



Flow: raw data lands in S3 → Glue catalogs and transforms it → Redshift/Athena serve it for analytics → insights drive decisions.

- Amazon S3 acts as durable, cost-effective cloud storage for raw and processed data.
- AWS Glue handles data cataloging, ETL, cleansing, and schema discovery.
- Redshift and Athena together provide fast, scalable querying and analytics on structured data.

Methodology

An 8-week, course-based structured learning plan

W1

Data Engineering for Data-Driven Orgs

Pipelines, the data engineer's role, hands-on with Amazon S3

W2

Data Elements & Pipeline Design

The 5 Vs of data, AWS Well-Architected Framework, Athena querying

W3

Securing, Scaling & Ingesting Pipelines

Cloud security, ETL vs ELT, data wrangling & cleaning

W4

Efficient Data Management & Big Data

Batch vs stream ingestion, AWS Glue & Kinesis, Redshift hands-on

W5

Big Data & Prepping Data for ML

Hadoop, Spark, EMR, and the SageMaker ML lifecycle

W6

Analyzing & Visualizing Data

IoT Analytics, QuickSight, Kinesis Data Firehose

W7

Automating the Pipeline

Step Functions, CI/CD, ETL pipeline labs with Athena

W8

Bridging to Certification

AWS certification tracks, benefits, and preparation resources

Expected Outcomes & Conclusion

A strong foundation in cloud-based data engineering



Fundamentals of AWS and cloud computing



Data Engineering concepts and workflows



Cloud-based data storage and processing



ETL and data transformation knowledge



Data pipelines and data integration basics



Preparing data for analytics and decisions

Overall, the internship strengthened technical knowledge of AWS and cloud-based data engineering, connecting academic learning with industry-oriented practice and building a foundation for further study in Cloud Computing and Data Engineering.